Advanced IO

System Input/Output

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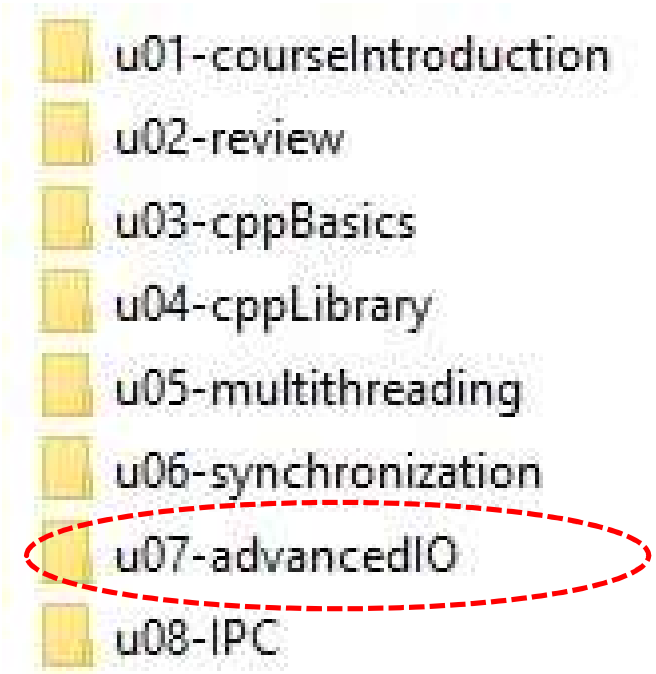
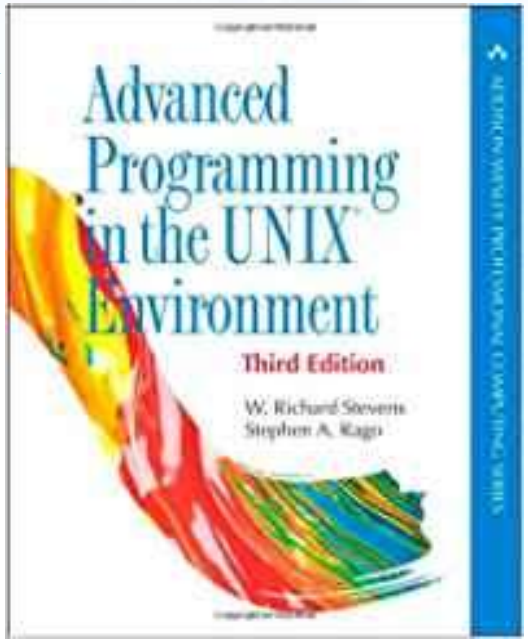
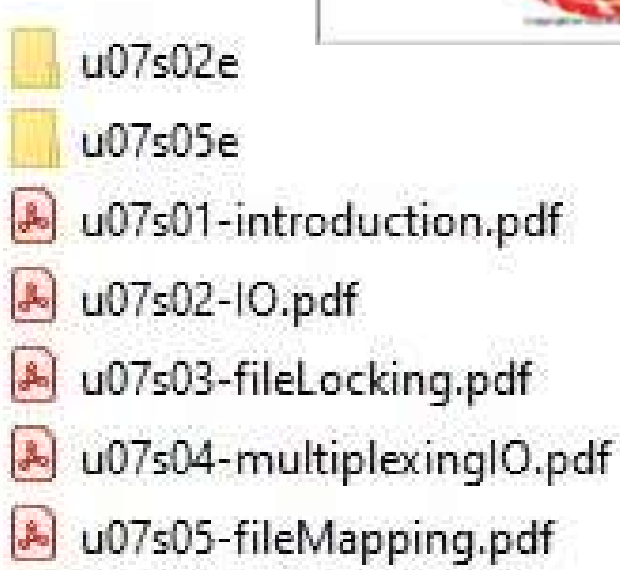
Premises

- Where are we?



Main reference





System calls

- In modern OSs hardware-related services are performed using system calls
  - Access a hard disk drive or a device, create and execute a new process, etc.
- System calls are how a program requests a service from the OS
  - They provide an essential interface between a process and the operating system
  - They are well-defined and limited in number
    - Linux and OpenBSD about 300
    - NetBSD and FreeBSD about 500
    - Windows about 2000

For more details, see the material of the OS course and Prof. Cabodi's part

System calls

- System calls are similar to functions but
  - Require the intervention of the kernel, i.e., a transfer of control from the user space to the kernel space
  - Need some sort of architecture-specific features like a software interrupt or a trap
  - They are by nature slow (as they require the intervention of the kernel)
    - They are usually divided into **slow** and **all the others** system calls

IO system calls

- Slow system calls are the ones that can block forever
  - Many are related to IO operations
    - **Opens** can block until some condition occurs on certain files
      - Until the unit is attached to the system (e.g., a file)
      - Until another process has the unit open for reading (e.g., a FIFO)
    - **Reads** can block if the data is not present
      - Read from stdin or a pipe
    - **Writes** can block if they cannot be accepted
      - A write to a full disk or a full pipe a write on a **locked** record in a file

See section u07s03

Synchronous system calls

- Moreover, standard IO system calls are usually synchronous
  - The task waits until the operation completes
  - Synchronous operations are inherently **slow** compared to other processing
  - Delays may be caused by
    - Hardware devices, e.g., track and sector, seek time on random access, etc.
    - Data transfer between a physical device and the system memory, etc.
    - Network transfer between file servers and computing devices, etc.

IO Efficiency

- In many applications, IO is fundamental and being inefficient implies enourmous time penalty
- The standard UNIX-POSIX introduces numerous advanced IO functionalities such as
  - Non-blocking and asynchronous IO
  - File locking
  - Multiplexing IO
  - Memory-mapped IO